Retrieval Quality as the Primary Failure Point in RAG-Based and Agentic Research Assistants

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Retrieval-Augmented Generation (RAG) has become a central architecture for grounding large language models (LLMs) in external knowledge, while increasingly agentic research assistants extend RAG with iterative search, planning, tool use, source evaluation, and multi-step reasoning. Despite substantial progress in generation quality, a fundamental systems problem remains: an assistant cannot reliably reason over evidence that it fails to retrieve, retrieves incompletely, or retrieves evidence that is relevant only superficially. This survey argues that retrieval quality should be treated as a primary upstream reliability concern in RAG-based and agentic research assistants rather than merely as one component among several. We synthesize research on information retrieval, dense retrieval, RAG, query expansion, long-context language modeling, corrective and self-reflective RAG, RAG evaluation, web agents, and emerging deep-research systems. We organize the literature around an evidence-centric taxonomy covering query formulation, corpus coverage, candidate retrieval, ranking, evidence granularity, contradiction coverage, source authority, temporal validity, provenance, and evidence utilization. We then compare existing approaches by retrieval strategy, adaptivity, evaluation setting, evidence handling, and limitations. Based on the synthesis, we propose a conceptual Retrieval Reliability Pipeline that moves from information-need analysis through multi-representation querying, high-recall retrieval, evidence-aware reranking, verification, adaptive retrieval, and evidence-constrained generation. The survey identifies open problems in claim-centric retrieval, retrieval uncertainty, contradiction-aware search, provenance graphs, retrieval-aware agent planning, and causal benchmarks for retrieval-induced failure. The central conclusion is deliberately nuanced: retrieval is not the only source of error, because models can fail to use or reason over correctly retrieved evidence, but inadequate retrieval can constrain every downstream stage and is therefore a critical target for reliability research.

CCS Concepts: • **Information systems** → **Information retrieval**; • **Computing methodologies** → *Artificial intelligence.*

Additional Key Words and Phrases: retrieval-augmented generation, RAG, agentic AI, research assistants, information retrieval, dense retrieval, query expansion, reranking, evidence grounding, retrieval evaluation, long-context language models, deep research

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1 Introduction

Large language models have transformed natural-language interfaces for search, question answering, summarization, coding, and scientific research. However, their knowledge is constrained by information encoded in model parameters and by the temporal and epistemic limitations of their training data. Retrieval-Augmented Generation (RAG) was introduced as a way of combining parametric language-model knowledge with non-parametric external memory, allowing a generator to condition its response on retrieved documents [7].

Modern research assistants have evolved beyond the relatively static pipeline query–retrieve–generate. Contemporary systems increasingly perform query reformulation, multi-hop retrieval,

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web search, source filtering, document analysis, iterative reasoning, tool invocation, and self-reflection. Systems such as STORM, Self-RAG, Search-o1, DeepResearcher, and Agent Laboratory illustrate this movement toward increasingly agentic research workflows [1, 8, 14, 15, 20].

This evolution creates a central systems question: *where does a RAG or agentic research assistant fundamentally fail?* Generation failures are visible because an answer may hallucinate, contradict retrieved documents, or make unsupported claims. Yet many downstream failures can originate earlier. If a system does not retrieve the relevant paper, retrieves only an abstract rather than the decisive experimental result, ranks a less authoritative source above a primary source, fails to retrieve contradictory evidence, or stops searching prematurely, a capable generator remains constrained by an inadequate evidence set.

This survey develops the thesis that retrieval quality is frequently an important upstream failure point in RAG-based and agentic research assistants. The word "primary" is used in a causal and systems sense, not as a claim that every observed error is caused by retrieval. Long-context research demonstrates that models can fail even when relevant information is present in the context [10]. Likewise, generation can fail through misinterpretation, unsupported inference, or reasoning errors. The stronger and more defensible proposition is that retrieval is a particularly consequential bottleneck because missing evidence cannot reliably be reconstructed by later stages.

The survey makes five contributions:

(1) It organizes retrieval quality around *evidence sufficiency* rather than document similarity alone.
(2) It develops a taxonomy of retrieval failure modes relevant to research-oriented RAG and agentic systems.
(3) It compares retrieval, RAG, evaluation, and agentic research approaches using common dimensions.
(4) It synthesizes research gaps and proposes a conceptual Retrieval Reliability Pipeline for adaptive, evidence-aware assistants.
(5) It identifies benchmark and methodological requirements for separating retrieval-caused errors from context-utilization and generation errors.

The remainder of the paper is organized as follows. Section 2 introduces conceptual foundations. Section 3 describes the literature-synthesis approach. Section 5 develops the retrieval-failure taxonomy. Section 6 compares existing retrieval and RAG approaches. Section 7 discusses agentic research assistants. Section 8 examines evaluation and benchmarks. Section 9 provides cross-study synthesis. Section 10 presents the proposed reliability framework. Sections 11 and 12 discuss limitations, gaps, and future directions, followed by the conclusion.

2 Background and Conceptual Foundations

2.1 From Information Retrieval to RAG

Classical information retrieval separates a user query from a corpus of candidate documents and ranks documents according to estimated relevance. Lexical methods such as BM25 remain important because exact terminology, rare entities, identifiers, and domain-specific vocabulary can be highly informative. Neural retrieval introduced dense vector representations, enabling semantic matching beyond literal lexical overlap. Dense Passage Retrieval (DPR) demonstrated the effectiveness of dual-encoder retrieval for open-domain question answering [6]. The BEIR benchmark subsequently showed that retrieval effectiveness varies substantially across heterogeneous domains and tasks, with BM25 remaining a strong zero-shot baseline in many settings [16].

RAG integrates retrieval with language generation. In the formulation of Lewis et al. [7], retrieved passages act as non-parametric memory that complements parametric model knowledge. This

architecture provides two important capabilities: external information can be incorporated without retraining the entire generator, and retrieved evidence can provide a basis for attribution.

A useful conceptual factorization is

$$P(\text{correct answer}) \approx P(\text{sufficient evidence retrieved})P(\text{evidence correctly used} \mid \text{retrieved})P(\text{generation corr} \tag{1}$$

Equation (1) is not intended as a complete probabilistic model. It expresses a structural dependency: if sufficient evidence is absent, later stages have limited ability to recover information that was never made available.

2.2 Retrieval Is More Than Similarity Search

Modern RAG systems commonly contain query formulation, candidate retrieval, hybrid lexical/dense retrieval, reranking, deduplication, diversification, context compression, evidence aggregation, and generation. Query2doc uses LLM-generated pseudo-documents for query expansion [17], while HyDE addresses semantic mismatch by generating a hypothetical document whose representation is used for retrieval [4]. These approaches demonstrate that retrieval quality depends partly on the representation of the information need, not only on the retriever architecture.

2.3 RAG and Long-Context Models

Long context does not eliminate the need for reliable information access. Liu et al. [10] showed that language models can use information less reliably when relevant content is located in the middle of long contexts. Thus three related failure classes should be distinguished: retrieval failure, context-utilization failure, and generation failure. A comparative study of RAG and long-context approaches also suggests that the relative advantage depends on task, model, and cost conditions [9].

3 Review Methodology

This manuscript is an enhanced scholarly synthesis of the previously developed review paper. The existing evidence base was retained as the primary source set, and the literature was reorganized thematically rather than summarized paper-by-paper. The source set spans foundational retrieval and RAG work, influential evaluation studies, and recent work on agentic research systems.

3.1 Literature Scope

The review focuses on research addressing retrieval effectiveness for knowledge-intensive question answering; query formulation and expansion; RAG retrieval, correction, reflection, or ranking; evaluation and diagnosis of retrieval or RAG failures; long-context evidence utilization; and agentic web search and research assistants.

3.2 Synthesis Dimensions

For comparative analysis, studies were organized using the dimensions shown in Table 1.

4 Retrieval Quality as an Evidence Bottleneck

A research assistant ultimately produces claims. Each externally verifiable claim should ideally be supported by evidence. Let the task require evidence units $E^* = \{e_1, e_2, \ldots, e_n\}$ while retrieval produces $R = \{r_1, r_2, \ldots, r_k\}$. The desired condition is not merely semantic similarity but sufficient evidence coverage:

$$E^* \subseteq \text{coverage}(R). \tag{2}$$

Table 1. Dimensions used to synthesize the reviewed literature.

Dimension	Question asked	Relevance to research assistants
Retrieval strategy	How are candidates generated?	Determines recall and corpus access.
Ranking	How are candidates prioritized?	Determines which evidence enters context.
Adaptivity	Can the system retrieve again?	Important for incomplete or multi-hop tasks.
Evidence granularity	What is the retrieval unit?	Determines whether exact claim support is available.
Source quality	Is authority modeled?	Important for scholarly and high-stakes research.
Temporal validity	Is freshness considered?	Important for dynamic information.
Contradiction handling	Is disagreement searched for?	Reduces confirmation-oriented synthesis.
Evaluation	Are retrieval and generation diagnosed separately?	Enables causal error analysis.

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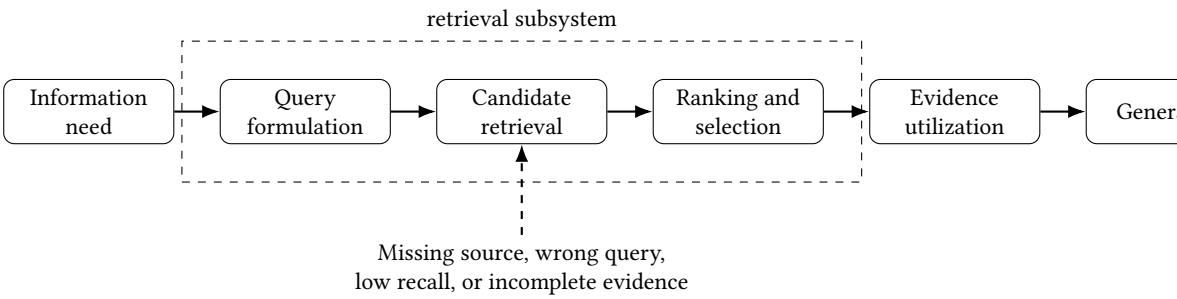


Fig. 1. Conceptual propagation of retrieval failures into downstream evidence utilization and generation.

For scholarly research questions, evidence can be distributed across multiple papers, methodological descriptions, datasets, tables, supplementary material, and contradictory findings. A retriever can therefore return several highly similar documents while omitting the one study containing the decisive result. Accordingly, retrieval quality should include recall and precision but also coverage, authority, temporal validity, diversity, consistency, granularity, and provenance.

5 Taxonomy of Retrieval Failure Modes

5.1 Query–Document Vocabulary Mismatch

A user may express an information need using conceptual language while relevant literature uses different terminology. Dense retrieval reduces lexical mismatch but does not eliminate semantic ambiguity. Query expansion can help, but an LLM-generated expansion may itself encode an incorrect interpretation [4, 17].

5.2 Corpus Coverage Failure

A retriever can only return information contained in its accessible corpus. Fixed vector databases can omit recent proceedings, preprints, supplementary material, specialized databases, or negative findings. Agentic systems operating on the open web face a different problem: the corpus is broad but noisy, dynamic, duplicated, and partially inaccessible [20].

5.3 Top-k and Ranking Failure

A common RAG system retrieves the top k documents and passes them to the generator. The relevant document may be ranked below k. Increasing k can increase noise, while reranking can improve precision at additional computational cost. RankRAG explores instruction-tuned ranking and generation [19].

5.4 Evidence Granularity Failure

A document may be relevant without containing the exact evidence required. A paper's abstract may establish that a method exists, but not whether it outperforms a baseline under a particular experimental condition. This motivates a document-to-section-to-paragraph-to-table-to-claim view of retrieval.

5.5 Redundancy and Correlated Evidence

Ten passages do not necessarily represent ten independent pieces of evidence. Preprints, conference versions, surveys, and web pages may repeat the same underlying study. Research assistants should therefore model source independence rather than simply count retrieved documents.

5.6 Contradictory Evidence

Research questions often require synthesis of disagreement. Retrieval optimized only for similarity can over-retrieve supporting evidence and miss dissenting or negative results. Research-oriented systems should therefore include deliberate counter-evidence search.

5.7 Temporal and Authority Failure

Research assistants can retrieve information that is relevant but outdated, secondary, or less authoritative than an available primary source. Freshness and source authority should therefore be explicit retrieval dimensions.

6 Existing Retrieval and RAG Approaches

6.1 Lexical and Dense Retrieval

BM25 remains a useful lexical baseline because exact terms can be decisive in technical domains. DPR demonstrated the value of dense representations for open-domain QA [6]. BEIR showed that no single retrieval method dominates uniformly across heterogeneous datasets [16]. This motivates hybrid systems that exploit complementary lexical and semantic signals.

6.2 Query Expansion and Multi-Representation Retrieval

Query2doc and HyDE show how LLMs can transform a query before retrieval [4, 17]. The benefit is improved semantic alignment, but the transformation itself is a possible failure point. An evidence-aware assistant should therefore maintain multiple query representations rather than trusting one generated reformulation.

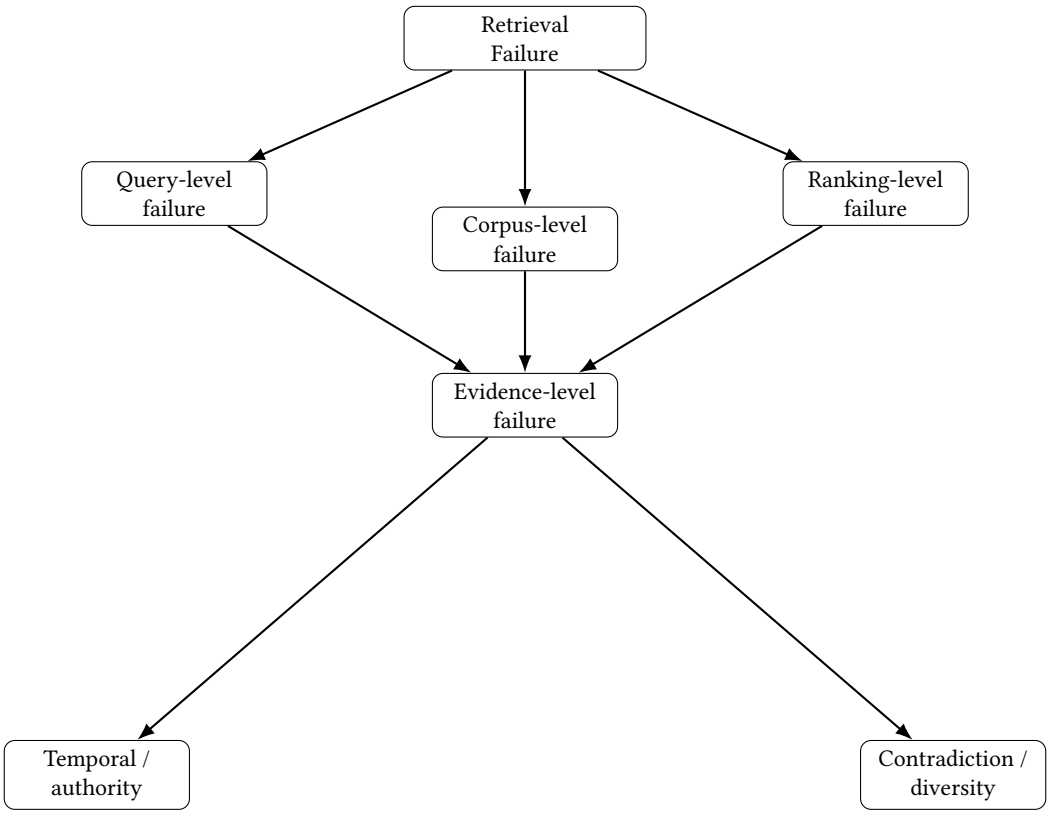


Fig. 2. Taxonomy of retrieval failures in research-oriented RAG and agentic assistants.

6.3 Reranking

Reranking separates high-recall candidate generation from more expensive relevance estimation. RankRAG demonstrates a unified ranking-and-generation direction [19]. Reranking can improve the quality of final context but increases latency and computation, making selective use important for long research tasks.

6.4 Corrective and Self-Reflective RAG

Corrective RAG introduces retrieval evaluation and corrective actions when retrieved evidence is inadequate [18]. Self-RAG allows a model to retrieve when necessary and reflect on retrieved passages and generated content [1]. Together, these approaches mark a conceptual transition from "retrieve once" to "retrieve, evaluate, correct, and retrieve again."

6.5 Evaluation-Aware RAG

RAGAs, ARES, RAGChecker, and RAGTruth demonstrate the movement toward multidimensional evaluation [3, 11–13].

7 Agentic Research Assistants

Agentic research assistants make retrieval a sequential decision process. Instead of a single query, an agent may formulate subquestions, search, inspect documents, revise its plan, search again, and

Table 2. Taxonomy of retrieval failure modes and mitigation directions.

Failure mode	Typical cause and effect	Possible mitigation
Query mismatch	User terminology differs from source terminology; relevant evidence is missed.	Query rewriting, multi-representation queries, lexical+dense retrieval.
Corpus incompleteness	Required source is absent from the accessible corpus.	Multi-source retrieval, web search, corpus expansion.
Top-k failure	Relevant evidence is ranked below the context cutoff.	High-recall candidate retrieval and reranking.
Ranking failure	A superficially relevant source displaces a decisive source.	Cross-encoder or LLM reranking; authority-aware ranking.
Granularity failure	Document is relevant but decisive evidence is in another section/table.	Passage-, section-, table-, or claim-level retrieval.
Redundancy	Multiple sources repeat the same underlying evidence.	Deduplication and source-independence modeling.
Contradiction failure	Supporting evidence is retrieved while disagreement is omitted.	Counter-evidence queries and diversity objectives.
Temporal failure	Outdated information is ranked above current evidence.	Freshness-aware retrieval and timestamp validation.
Authority/provenance failure bottomrule	Secondary or low-quality sources dominate.	Provenance graphs and primary-source preference.

synthesize evidence. STORM illustrates structured long-form research with source discovery and organization [15]. Search-o1 integrates search with reasoning for knowledge-intensive tasks [8]. DeepResearcher moves toward end-to-end training of research agents in realistic web environments and emphasizes planning, cross-validation, self-reflection, and redirection [20]. Agent Laboratory demonstrates a broader workflow in which agents assist with literature review, experimentation, and research reporting [14].

Web-oriented benchmarks reinforce the difficulty of realistic agentic environments. WebArena provides a realistic environment for autonomous web interaction [21], while WebVoyager investigates multimodal web agents operating on real websites [5]. Mind2Web provides a large-scale benchmark for generalist web agents [2].

Agentic retrieval introduces a sequential dependency:

$$a_1 \rightarrow r_1 \rightarrow a_2 \rightarrow r_2 \rightarrow \cdots \rightarrow a_n, \tag{3}$$

where a_i denotes an agent action and r_i denotes retrieved evidence. Equation (3) illustrates why an early retrieval error can influence later planning decisions.

8 Evaluation and Benchmarks

A major weakness of RAG evaluation is the tendency to evaluate only final answers. An incorrect answer may originate from query formulation, retrieval failure, ranking failure, incomplete evidence, contradictory sources, evidence-utilization failure, hallucination, or citation misalignment. A single end-to-end score cannot reliably identify the responsible component.

RAGAs separates retrieval relevance, faithfulness, and generation-related dimensions [3]. ARES evaluates context relevance, answer faithfulness, and answer relevance using automated judges with

Table 3. Comparison of representative retrieval and RAG approaches.

Approach	Core mechanism	Adaptivity	Main contribution	Main limitation
BM25	Lexical ranking	Low	Strong exact-term baseline.	Vocabulary mismatch.
DPR	Dense dual-encoder retrieval	Low	Semantic retrieval at scale.	Domain and representation sensitivity.
Query2doc	LLM-generated pseudo-document expansion	Medium	Improves query-document alignment.	Expansion can misrepresent information need.
HyDE	Hypothetical document embedding	Medium	Bridges query/document semantic gap.	Generated hypothesis may introduce noise.
RankRAG	Instruction-tuned ranking and generation	Medium	Richer context ranking.	Higher model and compute requirements.
CRAG	Retrieval evaluation plus corrective retrieval	High	Makes retrieval quality an explicit decision.	Additional retrieval and search cost.
Self-RAG	Retrieval plus reflection/critique	High	Joint retrieval and self-reflection.	Training and inference complexity.

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human annotations [13]. RAGChecker provides fine-grained diagnostics for retrieval and generation [12]. RAGTruth supplies a hallucination corpus with fine-grained annotations for RAG responses [11]. BEIR remains important for heterogeneous retrieval evaluation [16], while WebArena and WebVoyager expose difficulties in realistic interactive environments [5, 21].

A conceptual evaluation vector is

$$\mathcal{E} = (R, P, C, A, T, D, V, U, G), \tag{4}$$

where R is retrieval recall, P precision, C evidence coverage, A source authority, T temporal validity, D source diversity, V evidence verification, U evidence utilization, and G generation quality.

9 Comparative Synthesis of the Literature

The literature reveals a progression from static retrieval toward adaptive evidence management. Table 6 compares the approaches using a common set of dimensions. The comparison is descriptive rather than a ranking: each approach optimizes different parts of the retrieval-to-generation pipeline.

Three broad trends emerge. First, retrieval objectives are expanding from relevance toward evidence sufficiency. Second, retrieval is becoming adaptive rather than fixed. Third, evaluation is becoming increasingly diagnostic. These trends are mutually reinforcing: once retrieval is treated as a decision process, evaluation must determine whether a retrieval decision reduced uncertainty; once evidence completeness is emphasized, a single top-k score becomes insufficient.

Table 4. Representative agentic research and web-agent systems.

System	Primary setting	Retrieval behavior	Contribution	Limitation relevant to retrieval
STORM	Long-form research	Iterative source exploration	Structured research and synthesis.	Source selection and synthesis remain dependent on search quality.
Search-o1	Agentic search	Search integrated with reasoning	Connects information seeking with reasoning.	Sequential search increases latency and error propagation.
DeepResearcher	Open web	Dynamic multi-agent browsing	RL-trained deep research in realistic web environments.	Open-web noise and dynamic pages remain difficult.
Agent Laboratory	End-to-end research	Literature search plus research workflow	Demonstrates broader research automation.	Human oversight remains important.
WebArena	Web benchmark	Multi-step website interaction	Realistic evaluation environment.	Task success mixes retrieval with action and interface errors.
WebVoyager	Multimodal web agent	Visual and textual web navigation	End-to-end web-agent capability.	Web interaction introduces additional non-retrieval failures.

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10 Retrieval Reliability Framework

Based on the synthesis, this survey proposes a conceptual Retrieval Reliability Pipeline (RRP). The framework does not claim a validated experimental improvement; it organizes requirements emerging from the reviewed literature.

10.1 Stage 1: Information-Need Analysis

The assistant identifies entities, constraints, temporal requirements, methodological requirements, subquestions, and expected evidence types.

10.2 Stage 2: Multi-Representation Querying

The system maintains the original query, lexical query, semantic query, decomposed queries, query expansions, and counter-evidence queries.

Table 5. Representative evaluation resources and what they diagnose.

Resource	Evaluation focus	Useful diagnostic role	Important limitation
BEIR	Heterogeneous information retrieval	Tests generalization across domains and tasks.	Does not directly measure downstream answer reliability.
RAGAs	RAG evaluation	Separates retrieval relevance and generation-related dimensions.	Automated metrics can inherit judge/model biases.
ARES	Context and answer evaluation	Measures context relevance, faithfulness, and answer relevance.	Requires careful calibration and annotation.
RAGChecker	Fine-grained RAG diagnosis	Helps identify retrieval versus generation problems.	Diagnostic granularity increases evaluation complexity.
RAGTruth	Hallucination analysis	Provides fine-grained hallucination annotations.	Focus is narrower than complete retrieval reliability.
WebArena	Web-agent interaction	Tests long-horizon autonomous interaction.	Overall success mixes retrieval with action and interface errors.
WebVoyager	Multimodal web agents	Evaluates realistic web navigation.	Retrieval is only one component of task difficulty.
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10.3 Stage 3: High-Recall Retrieval

Multiple retrieval strategies can be combined, including BM25, dense retrieval, hybrid retrieval, scholarly search, and open-web retrieval.

10.4 Stage 4: Evidence-Aware Reranking

Candidates should be ranked according to relevance, evidence coverage, authority, freshness, and diversity rather than similarity alone.

10.5 Stage 5: Evidence Verification

The system checks whether a passage actually supports the intended claim, whether the source is primary, whether conflicting evidence exists, and whether the source is sufficiently current.

10.6 Stage 6: Adaptive Retrieval

If evidence is insufficient, the system identifies the missing evidence requirement, retrieves again, evaluates the new evidence, and repeats.

10.7 Stage 7: Evidence-Constrained Generation

The generator distinguishes supported claims, model inference, and unresolved uncertainty. Figure 3 summarizes the complete pipeline.

Table 6. Comparative synthesis of representative research approaches.

Approach family	Retrieval model	Adaptivity	Evidence handling	Typical evaluation	Key limitation
Classical IR	Lexical	Low	Document relevance	IR benchmarks	Vocabulary mismatch.
Dense retrieval	Neural semantic	Low	Passage relevance	QA/IR benchmarks	Domain shift.
Query expansion	LLM-assisted	Medium	Query representation	Ad-hoc retrieval	Expansion error.
Reranking	Cross-encoder/LLM	Medium	Candidate prioritization	Retrieval/RAG benchmarks	Latency and compute.
Corrective RAG	Evaluator + retrieval	High	Retrieval quality checking	RAG tasks	Additional search cost.
Self-reflective RAG	Retrieval + critique	High	Passage and generation reflection	Knowledge-intensive tasks	Complexity and training requirements.
Agentic research	Iterative web/search tools	High	Multi-step evidence gathering	Research/web environments	Error compounding and noisy sources.

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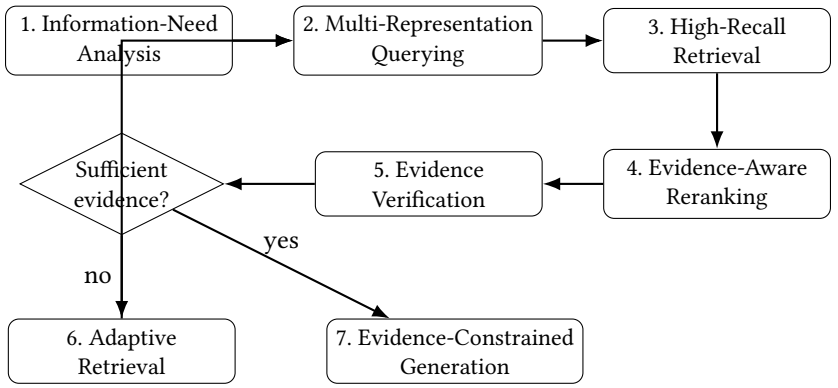


Fig. 3. Conceptual Retrieval Reliability Pipeline (RRP) proposed from the literature synthesis.

11 Research Challenges and Limitations

11.1 The Primary-Failure Claim Is Not Universal

The thesis should not be interpreted as claiming that retrieval always causes the largest proportion of errors. Long-context research shows that models may fail even when relevant information is present [10]. The defensible claim is that retrieval is frequently an upstream bottleneck whose failure constrains downstream reasoning.

11.2 Retrieval Metrics Do Not Equal Research Quality

High Recall@k does not guarantee a useful research answer. A system can retrieve the correct document while failing because the relevant section is omitted, the source is outdated, contradictory findings are absent, or the evidence does not answer the complete question.

11.3 Dynamic Web Retrieval Is Noisy

Open-web research introduces duplicated content, low-quality pages, changing pages, unreliable metadata, inaccessible documents, and adversarial content. WebArena and WebVoyager illustrate why realistic web environments are substantially harder than simplified retrieval settings [5, 21].

11.4 Agentic Error Compounding

In an agentic loop, an early misleading retrieval can influence subsequent queries and planning. This creates a retrieval-induced reasoning loop in which the agent repeatedly gathers evidence consistent with an initial mistaken interpretation.

11.5 Evaluation and Attribution

Fine-grained evaluation is computationally and methodologically more demanding than a single end-to-end score. Future studies must balance diagnostic resolution with practical evaluation cost.

12 Research Gaps and Future Directions

12.1 From Relevance to Reliability

Current retrieval research often emphasizes relevance. Research assistants require a broader objective:

$$\text{Retrieval Reliability} = f(\text{relevance}, \text{coverage}, \text{authority}, \text{diversity}, \text{freshness}, \text{consistency}, \text{provenance}). \tag{5}$$

12.2 Claim-Centric Retrieval

Future systems should begin with claims or subquestions rather than documents. For a complex question Q, an agent can construct

$$Q \rightarrow \{q_1, q_2, \ldots, q_m\}, \tag{6}$$

and associate each subquestion with explicit evidence requirements.

12.3 Retrieval Uncertainty and Search Stopping

An agent must decide not only what to retrieve but when to stop. A stopping policy should consider evidence coverage, source diversity, authority, unresolved contradiction, and expected marginal information gain rather than a fixed document count.

12.4 Contradiction-Aware Retrieval

After retrieving evidence supporting a claim c, an assistant could explicitly issue a counter-evidence query such as "evidence against c." This turns research from confirmation-oriented retrieval into adversarial evidence gathering.

12.5 Provenance-Aware Research Graphs

Future assistants should represent evidence as Claim → Evidence → Source → Experiment → Dataset. Such graphs can help determine whether apparently independent sources ultimately depend on the same study.

Table 7. Research gaps and corresponding future research directions.

Research gap	Why existing evaluation is insufficient	Future direction
Evidence completeness	Relevance scores do not establish support for all required claims.	Claim-level evidence coverage benchmarks.
Retrieval uncertainty	Fixed top-k retrieval assumes a universal stopping rule.	Uncertainty- and information-gain-based stopping.
Contradiction coverage	Similarity retrieval can favor consensus or repeated claims.	Counter-evidence retrieval and diversity objectives.
Provenance	Document counts can overstate evidence independence.	Claim–source–experiment provenance graphs.
Temporal reliability	Static benchmarks underrepresent changing information.	Time-aware and dynamic retrieval benchmarks.
Causal attribution	End-to-end scores cannot isolate retrieval-caused errors.	Controlled retrieval interventions and causal evaluation.
Agentic error propagation	Sequential retrieval creates dependent failures.	Benchmarks for error compounding across search trajectories.

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12.6 Retrieval-Aware Agent Planning

Search should be treated as a planning action with expected information gain. An agent could choose among broad search, database-specific search, primary-paper retrieval, supplementary-material inspection, contradiction search, and stopping.

12.7 Benchmarks for Retrieval-Caused Failure

A major gap is the absence of benchmarks specifically designed to answer: did the assistant fail because it could not retrieve the required evidence, or because it failed to reason over evidence already present? Controlled interventions should vary evidence availability, ranking position, source authority, contradiction, freshness, and completeness.

13 Conclusion

RAG has changed the architecture of language-model systems by separating knowledge storage from language generation and enabling models to access external evidence. As these systems evolve into agentic research assistants, retrieval becomes increasingly central rather than peripheral.

The literature reviewed in this survey supports a nuanced conclusion. Retrieval is not the only source of failure: language models may fail to use retrieved evidence, generate unsupported claims, or reason incorrectly even when retrieval succeeds. However, retrieval remains a particularly important upstream bottleneck because missing or inadequate evidence constrains all subsequent stages. Research on dense retrieval, heterogeneous retrieval benchmarks, query expansion, corrective RAG, self-reflective RAG, long-context processing, automated RAG evaluation, and agentic research systems collectively reinforces this view.

The most important conceptual shift is from asking "Did the system retrieve a relevant document?" to asking "Did the system retrieve sufficient, authoritative, diverse, current, and verifiable evidence

to support the claims it ultimately makes?" For research assistants, relevance alone is not an adequate reliability criterion.

The proposed Retrieval Reliability Pipeline in Figure 3 operationalizes this shift through information-need analysis, multi-representation querying, high-recall retrieval, evidence-aware reranking, evidence verification, adaptive retrieval, and evidence-constrained generation. The framework is conceptual and should be validated experimentally rather than treated as an established result.

Future RAG research should therefore treat retrieval not as a preprocessing step but as a core epistemic component of intelligent systems. The most promising direction is not simply developing larger retrievers or longer contexts; it is developing retrieval-reliable agents whose search behavior is adaptive, evidence-aware, uncertainty-sensitive, provenance-preserving, and explicitly evaluated for retrieval-caused failure. Controlled retrieval interventions, claim-level evidence benchmarks, causal error attribution, and agentic research evaluations can provide the empirical basis for testing this proposition.

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